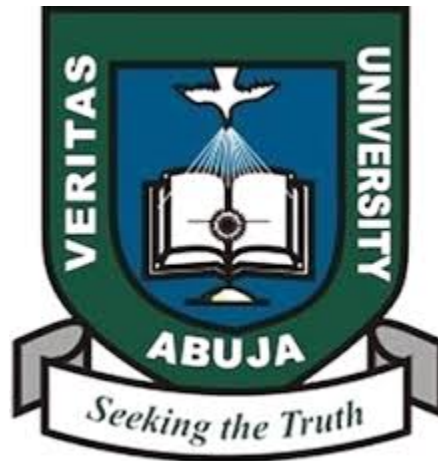


Department of Software Engineering
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‘LAB MANUAL’

Course Code: VUA-SEN182

Course Title: “Web Programming (HTML, CSS, Git)”

Prepared by: Mr. Victor Ayamba

Course Contents

Contents	CLOs	Lecture	Practical
HTML: Basics, Elements, Attributes, Headings, Paragraphs, Formatting, Links, Head, Images, Tables, Lists, Blocks, Layout, Forms, Multimedia Tags, Semantic HTML, HTML5 Structure.	CLO – 1 & 2	4	8
CSS: Introduction, Syntax, Selectors (ID & Class), Styling Backgrounds, Text, Fonts, Links, Box Model (Border, Outline, Margin, Padding), Grouping/Nesting, Dimension, Display, Positioning, Floating, Align, Navigation Bar, Image Gallery, Flexbox, Grid Layout, Responsive Design.	CLO – 1 & 2	3	6
JavaScript: Introduction, Output, Statements, Comments, Variables, Data Types, Objects, Functions, Operators, Comparisons, Conditions, Loops, Error Handling, DOM Introduction and Manipulation, Number, String, Date, Array, Window, Screen, Navigator, Popup Alerts.	CLO – 1 & 2	4	8
Bootstrap: Introduction to Bootstrap Framework, Grid System, Containers and Layout, Responsive Design Principles, Typography and Utilities, Navigation Bars, Buttons, Cards, Forms and Input Groups, Tables, Alerts, Modals, Responsive Components and Layout Customization.	CLO – 1 & 2	4	8
Total		15	30

INTRODUCTION TO THE COURSE:

The Internet and the World Wide Web have become fundamental components of modern society, influencing communication, education, business, and everyday life. As digital platforms continue to expand, the ability to design and develop functional and visually appealing web applications has become an essential skill for software engineers and technology professionals. Understanding how web technologies work together enables students to create interactive systems that can serve real-world needs.

This laboratory course provides a practical introduction to modern Web Programming using core client-side technologies such as **HTML, CSS, JavaScript, and the Bootstrap framework as well as version control tools like Git and Github**. Students will learn how to structure web pages using HTML, style and layout interfaces with CSS, enhance interactivity through JavaScript, and build responsive, mobile-friendly user interfaces using Bootstrap. Through hands-on laboratory exercises, students will progressively develop the skills required to design, implement, and maintain modern web applications.

The lab sessions are structured to guide students step-by-step through the development of a meaningful web project while introducing essential web development concepts and tools. Each experiment builds upon the previous one, enabling learners to gradually transform basic web pages into fully structured and responsive websites. At the end of each lab session, tasks are provided to reinforce understanding and encourage independent exploration. Students are advised to follow this lab manual sequentially and carefully adhere to the provided guidelines to gain maximum benefit from the course.

Lab 5: Introduction to Git and GitHub - Demonstrated by Instructor

Introduction to Version Control

Version control is a system that helps developers manage changes to files over time. It allows you to track modifications, revert to previous versions, and collaborate with others.

Git is a **distributed version control system**, meaning every user has a full copy of the project history.

GitHub is a **cloud-based platform** that hosts Git repositories and allows collaboration, sharing, and backup of projects.

What is Git?

Git is a tool that allows developers to:

- Track changes in code
- Save project history
- Collaborate with others
- Restore previous versions

Git works through commands executed in the **command line (CLI)**.

What is GitHub?

GitHub is a platform that allows you to:

- Store your Git repositories online
- Share your projects
- Collaborate with others
- Manage versions remotely

Installing Git

Step 1: Download Git

Visit: <https://git-scm.com>

Download Git based on your operating system.

Step 2: Install Git

Run the installer and follow default settings.

Step 3: Verify Installation

Open terminal or command prompt and run:

```
git --version
```

If installed correctly, it will display the Git version.

Installing GitHub Desktop (Optional)

Although this course uses CLI, GitHub Desktop can help beginners understand Git visually.

Download from: <https://desktop.github.com>

Creating a GitHub Account

1. Visit <https://github.com>
2. Click **Sign Up**
3. Enter username, email, and password
4. Verify your account

Configuring Git

Before using Git, set your identity:

```
git config --global user.name "Your Name"
git config --global user.email "your@email.com"
```

Check configuration:

```
git config --list
```

Creating a Local Repository

A repository (repo) is a folder where your project is stored and tracked.

Step:

- `mkdir orientation-portal`
- `cd orientation-portal`

- `git init`

This initializes a Git repository.

Checking Repository Status

`git status`

This shows:

- Untracked files
- Changes not yet committed

Adding Files to Git

To track files:

```
git add index.html
```

To add all files:

```
git add .
```

Committing Changes

A commit saves changes to the project history.

```
git commit -m "Initial commit of orientation portal"
```

Viewing Commit History

```
git log
```

Creating a GitHub Repository

1. Go to GitHub
2. Click **New Repository**
3. Enter repository name (e.g., orientation-portal)
4. Click **Create Repository**

Connecting Local Repo to GitHub

```
git remote add origin https://github.com/username/orientation-portal.git
```

Pushing Code to GitHub

```
git push -u origin main
```

Pulling Changes

```
git pull origin main
```

Cloning a Repository

```
git clone https://github.com/username/repository-name.git
```

Git Basic Commands Summary

Command	Description
git init	Initialize repository
git status	Check status
git add	Stage files
git commit	Save changes
git push	Upload to GitHub
git pull	Download updates
git clone	Copy repository

SECTION B: Lab Tasks on Git and GitHub

Task 1: Install Git

What to do

Download and install Git on your system.

Visit:

<https://git-scm.com>

Verify installation:

`git --version`

Task 2: Create a GitHub Account

What to do

Create a GitHub account.

Suggestion

Go to:

<https://github.com>

Sign up and verify your email.

Task 3: Configure Git

What to do

Set your username and email in Git.

Suggestion

```
git config --global user.name "Your Name"
```

```
git config --global user.email "your@email.com"
```

Task 4: Create Project Folder

What to do

Create a folder for your Orientation Portal project.

Task 5: Initialize Git Repository

What to do

Initialize Git inside your project folder.

Task 6: Add Your HTML Project

What to do

Copy your Lab 1–4 HTML files into the project folder.

Task 7: Stage Files

What to do

Add your files to Git staging area.

Suggestion

```
git add .
```

Task 8: Commit Your Work

What to do

Save your project in Git.

Suggestion

```
git commit -m "Added orientation portal project"
```

Task 9: Create GitHub Repository

What to do

Create a repository on GitHub.

Suggestion

Name it a name that reflects what you are working on; either our class name or something tangible

Task 10: Connect Local Repo to GitHub

What to do

Link your local project to GitHub.

Suggestion

```
git remote add origin https://github.com/your-username/orientation-portal.git
```

Task 11: Push Project to GitHub

What to do

Upload your project to GitHub.

Suggestion

```
git push -u origin main
```

Task 12: Make Changes and Update

What to do

Edit your HTML file (e.g., add a new section).

Suggestion

```
git add .  
git commit -m "Updated orientation page"  
git push
```

Task 13: Clone a Repository

What to do

Download a repository from GitHub via cli.

Task 14: Pull Updates

What to do

Get updates from GitHub by using `git pull origin main`

Task 15: Full Project Submission

What to do

Ensure your Orientation Portal project is fully uploaded to GitHub.

Expected Outcome

- Repository created

- Files uploaded
- At least **2 commits made**

Final Expected Outcome

At the end of this lab, students should be able to:

- ☐ Install Git
- ☐ Create GitHub account
- ☐ Initialize repository
- ☐ Track changes
- ☐ Commit code
- ☐ Push to GitHub
- ☐ Clone and pull repositories